



Customer Shopping Behavior Analysis

Uncovering insights from **3,900 transactions** to guide smarter business decisions.

The illustration depicts a professional in a blue suit working at a desk. The computer monitor shows a data dashboard with the title "Data Alaajist" and a date "205 TSATs". The dashboard includes several charts and graphs, all labeled in Armenian:

- Top Left:** "Rnars: Analylar" (Ranks: Analysis) with a legend for "Pisla" (Pink), "Indatibst" (Yellow), and "bnd" (Blue).
- Top Center:** "Dairis" (Circles) and "Elegi" (Selection) charts.
- Top Right:** "Dairis" (Circles) and "Dairis" (Circles) charts.
- Middle Left:** "Dairis: Retedat Ovtooyosor" (Circles: Retained Automaticity) chart.
- Middle Center:** "Dairis" (Circles) chart.
- Middle Right:** "Dairis" (Circles) chart.
- Bottom Left:** "Dmcrck: PoacTavest" (Check: PoacTavest) with a legend for "Ratijht" (Blue) and "Cvdrk Dorayd" (Orange).
- Bottom Center:** "Dairis" (Circles) chart.
- Bottom Right:** "Dairis" (Circles) chart.

The person is holding a pen and pointing at the screen, and has a calculator and papers on the desk.

Categories

Key Data Features



Demographics

Age, Gender, Location, Subscription Status



Shopping Behavior

Discounts, Promo Codes, Frequency, Review Rating



Purchase Details

Item, Category, Amount, Season, Size, Color



Payment & Shipping

Payment Method, Shipping Type

Data Preparation in Python



Load & Explore

pandas `df.info()` and `describe()`

Clean Missing Data

Imputed 37 Review Ratings via category median

Feature Engineering

Created `age_group` and `purchase_frequency_days`

Database Load

Cleaned DataFrame loaded into MySQL

SQL Business Questions

Revenue by Gender

**High-Spending Discount
Users**

Top 5 Products by Rating

Shipping Type Comparison

Subscribers vs. Non-Subscribers

More SQL Insights

01

Discount-Dependent Products

02

Customer Segmentation

New · Returning · Loyal

03

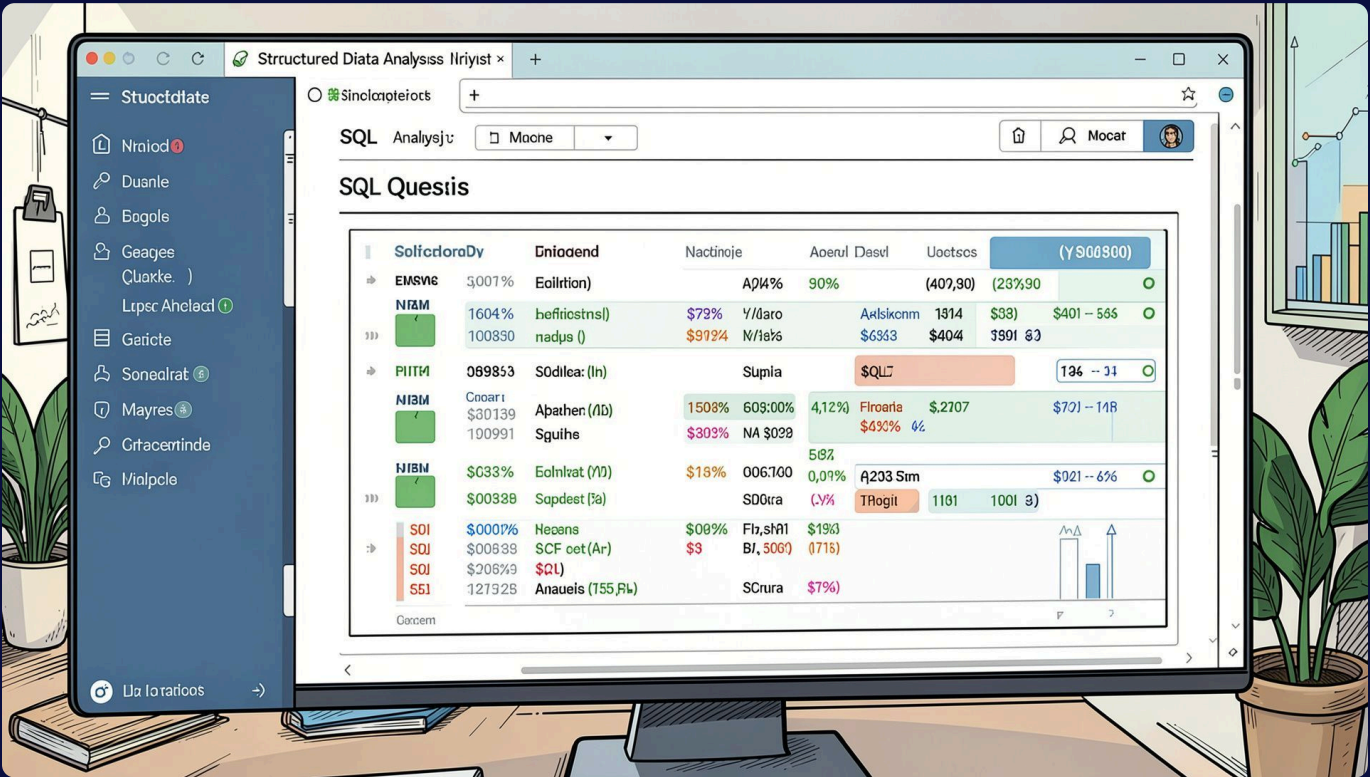
Top 3 Products per Category

04

Repeat Buyers & Subscriptions

05

Revenue by Age Group



Power BI Dashboard



Key Metrics

3.9K

Customers

\$59.76

Avg. Purchase

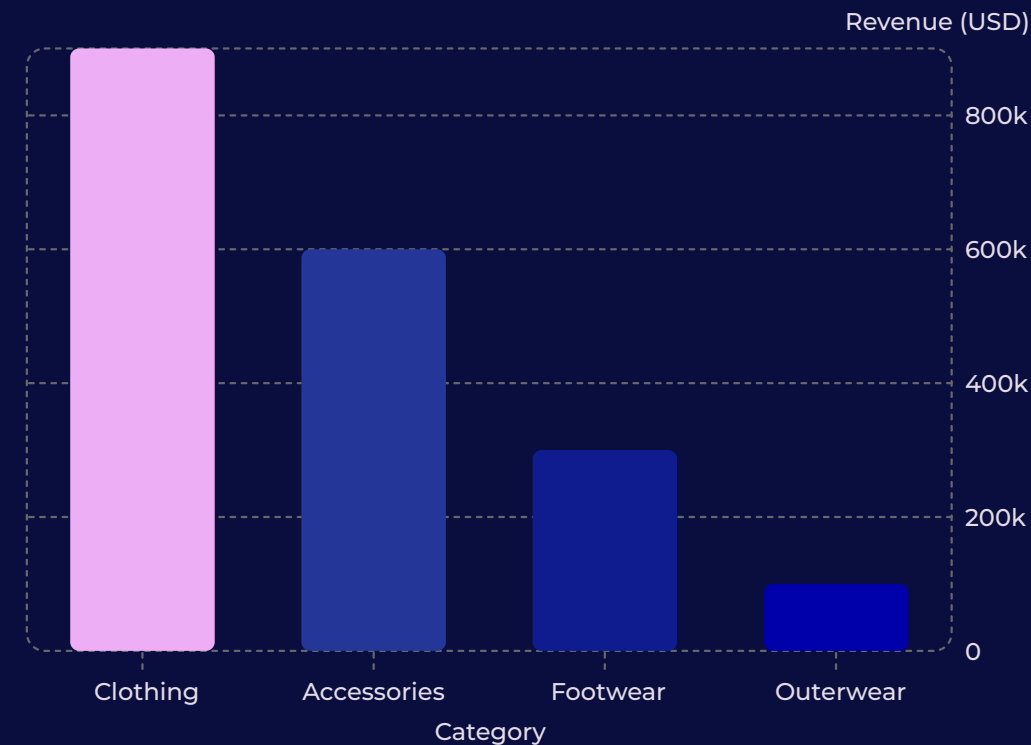
3.75

Avg. Rating

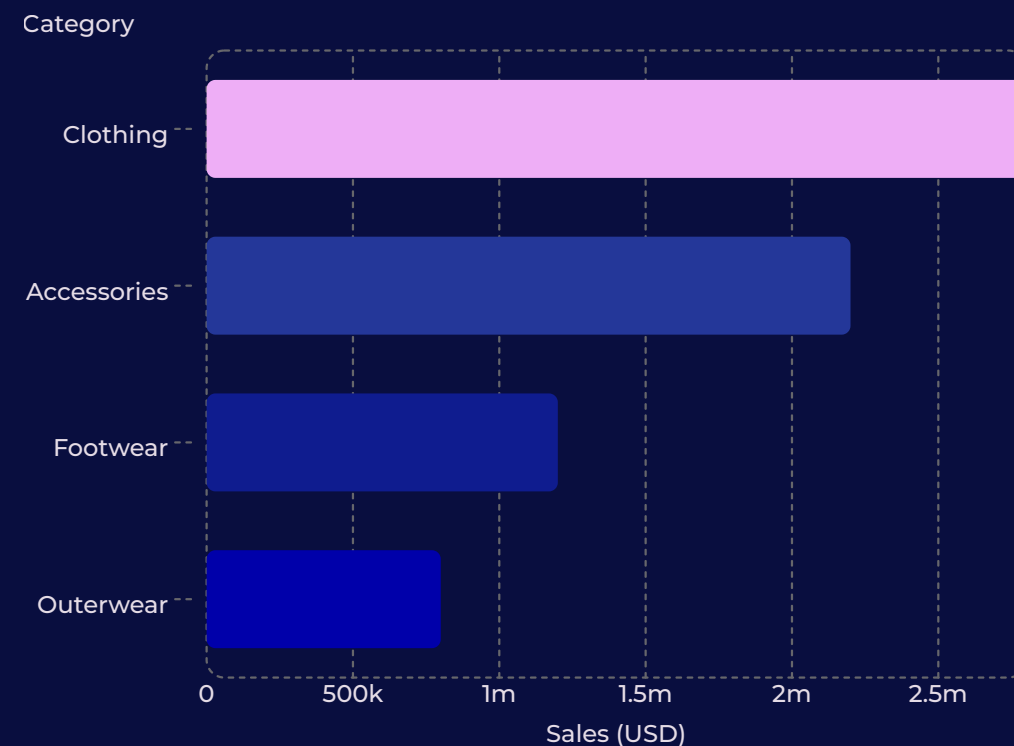
Filters

Subscription · Gender · Category ·
Shipping Type

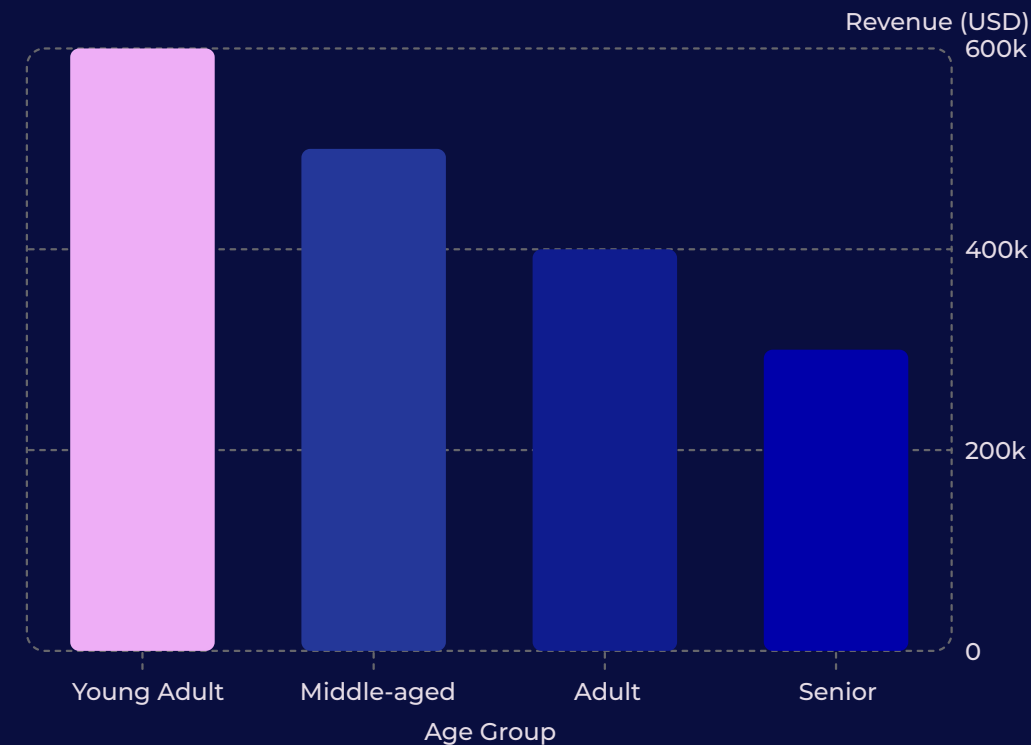
Revenue & Sales by Category



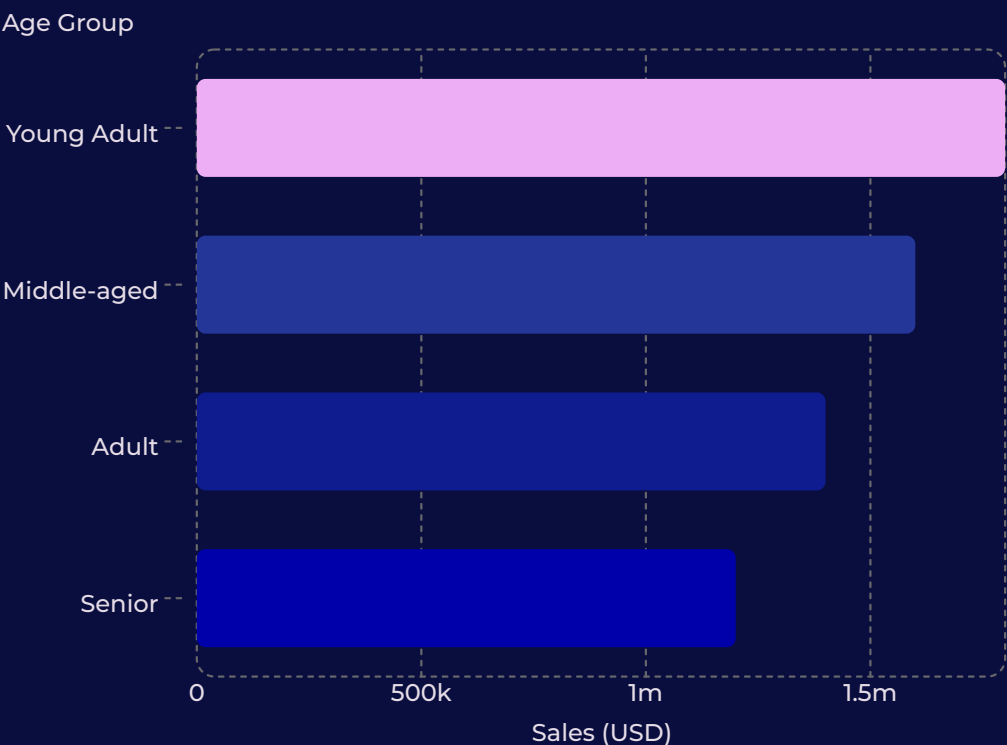
Sales by Category



Revenue & Sales by Age Group



Sales by Age Group



Business Recommendations



Boost Subscriptions

Promote exclusive subscriber benefits



Loyalty Programs

Reward repeat buyers toward "Loyal" segment



Review Discounts

Balance sales boosts with margin control



Targeted Marketing

Focus on high-revenue age groups & express-shipping users



Product Positioning

Highlight top-rated & best-selling products